

Same answers as `lista7_resolucao.pdf`, written the way the instructor's keys are: result first, arrow chains, derivation only where the answer needs it. Numbers: Benigno's calibration, $\alpha = 0,66$, $\sigma = 0,5$, $\eta = 0,2 \Rightarrow \kappa \approx 1,13$; $\theta = 8$. SR = short run, LR = long run.

Here σ is the elasticity of intertemporal substitution (scaled by C/Y); in Kurlat σ is curvature and $1/\sigma$ the EIS. Opposite conventions.

(1) NK vs New Classical (NC) vs original Keynesian (K)

Summary: NK = NC method (optimisation, rational expectations) + K friction (prices don't clear in the SR), plus monopolistic competition and an interest-rate instrument.

(i) Prices, expectations, monetary policy

- K: rigidity assumed, adaptive/no expectations $\rightarrow$ stable, exploitable Phillips curve $\rightarrow$ money has lasting real effects.
- NC: flexible prices, RE $\rightarrow$ only surprises move y (Lucas) $\rightarrow$ systematic policy ineffective (Sargent–Wallace).
- NK: RE + share α of prices pre-set at $p^e \rightarrow$ same equation $p - p^e = \kappa(y - y_n)$, but p^e is a *posted price*, not a forecast $\rightarrow$ known policy reacting after p^e still moves y ; LR neutral.
- Evidence: 1970s stagflation (keep RE); announced disinflations still cost output (need a friction); micro prices last ≈ 3 quarters ($\alpha \approx 0,66$).
- Implication: $\kappa = \frac{(1 - \alpha)(\sigma^{-1} + \eta)}{\alpha}$ is structural; $\alpha \uparrow \rightarrow \kappa \downarrow \rightarrow$ AS flatter.

Same curve as Lucas (Benigno p. 508) — not the forward-looking NKPC.

(ii) Market structure

- K: unmodelled / rigid nominal wage $\rightarrow$ countercyclical real wage (not in data).
- NC: perfect competition $\rightarrow$ nobody sets a price $\rightarrow$ nobody to be sticky.
- NK: Dixit–Stiglitz, $P(j) = \underbrace{\frac{\theta}{\theta-1}}_{\text{mark-up}} \frac{W}{A} \rightarrow$ (1) firms set prices; (2) $P > MC \rightarrow$ stuck firm serves extra demand $\rightarrow$ y demand-determined in the SR; (3) mark-up = wedge $\rightarrow y_n < y_e$.
- Menu cost: private loss 2nd order, aggregate effect 1st order (Mankiw 1985).

(iii) Aggregate demand and the instrument

- K: IS (consumption function) + LM (money demand); CB sets M ; slope via real balances.
- NC: $MV = PY$; M is the instrument and the nominal shock.
- NK: AD = Euler equation, $y = \bar{y}_n - \sigma[i - (\bar{p} - p) - \rho]$; CB sets i ; no LM.
Slope: $p \uparrow \rightarrow (\bar{p} - p) \downarrow \rightarrow r \uparrow \rightarrow C \downarrow \rightarrow y \downarrow$.
- Evidence: CBs target short rates, money demand unstable. Implication: transmission via r ; expectations $(\bar{y}_n, \bar{p})$ shift demand today; ZLB possible.

No LM curve: never explain AD's slope with money demand.

(iv) Microfoundations and welfare

- K: postulated equations, ad hoc loss. NC: micro-founded but efficient $\rightarrow$ nothing to stabilise.
- NK: micro-founded *with* distortions $\rightarrow$ loss *derived* from utility: $L = \frac{1}{2}(y - y_e)^2 + \frac{\theta}{2\kappa}(p - p^e)^2$
- Target y_e (not y_n); only surprises cost (price dispersion); weight $\theta/\kappa \uparrow$ with α, θ . Evidence: Lucas critique $\rightarrow$ deep parameters.

(2) AS-AD. Start: $i = r_n = \rho$, $y = y_n = y_e = 0$, $p = p^e = \bar{p}$. Closed form:

$$y - y_n = \frac{\sigma}{1 + \sigma\kappa} (r_n - i), \quad p - p^e = \kappa(y - y_n), \quad r_n = \rho + \sigma^{-1}(\bar{y}_n - y_n)$$

(a) **Temporary** $\mu \uparrow \Rightarrow$ AS shifts up/left; AD does not move

$\mu \uparrow \rightarrow dy_n = -\frac{d\mu}{\sigma^{-1} + \eta} < 0$, $dy_e = 0 \rightarrow$ AS redrawn through (p^e, y'_n) , vertical shift $\frac{1-\alpha}{\alpha}d\mu$.

AD: $\mu \notin$ AD and $\bar{\mu}$ unchanged $\rightarrow \bar{y}_n$ unchanged $\rightarrow$ AD fixed.

SR: firms want bigger margin $\rightarrow p \uparrow \rightarrow (\bar{p} - p) \downarrow \rightarrow r \uparrow \rightarrow C \downarrow \rightarrow y \downarrow$

$$dy = \frac{\sigma\kappa}{1 + \sigma\kappa} dy_n < 0, \quad dp = -\frac{\kappa}{1 + \sigma\kappa} dy_n > 0 \quad \text{stagflation}$$

$d\mu = 5\%$: $dy = -0,82\%$, $dp = +1,64\%$. Gaps: $y - y_n = +1,45\% > 0$ (why $p \uparrow$); $y - y_e = -0,82\% < 0$ (why welfare $\downarrow$).

LR: every firm resets $\rightarrow y = \bar{y}_n$, $p = \bar{p}$.

Two gaps, opposite signs. Demand falls along AD, AD does not shift. A permanent $\mu \uparrow$ would also lower $\bar{y}_n \rightarrow$ AD down.

(b) $i \downarrow \Rightarrow$ AD shifts up by $|di|$ (right by $\sigma|di|$); AS does not move

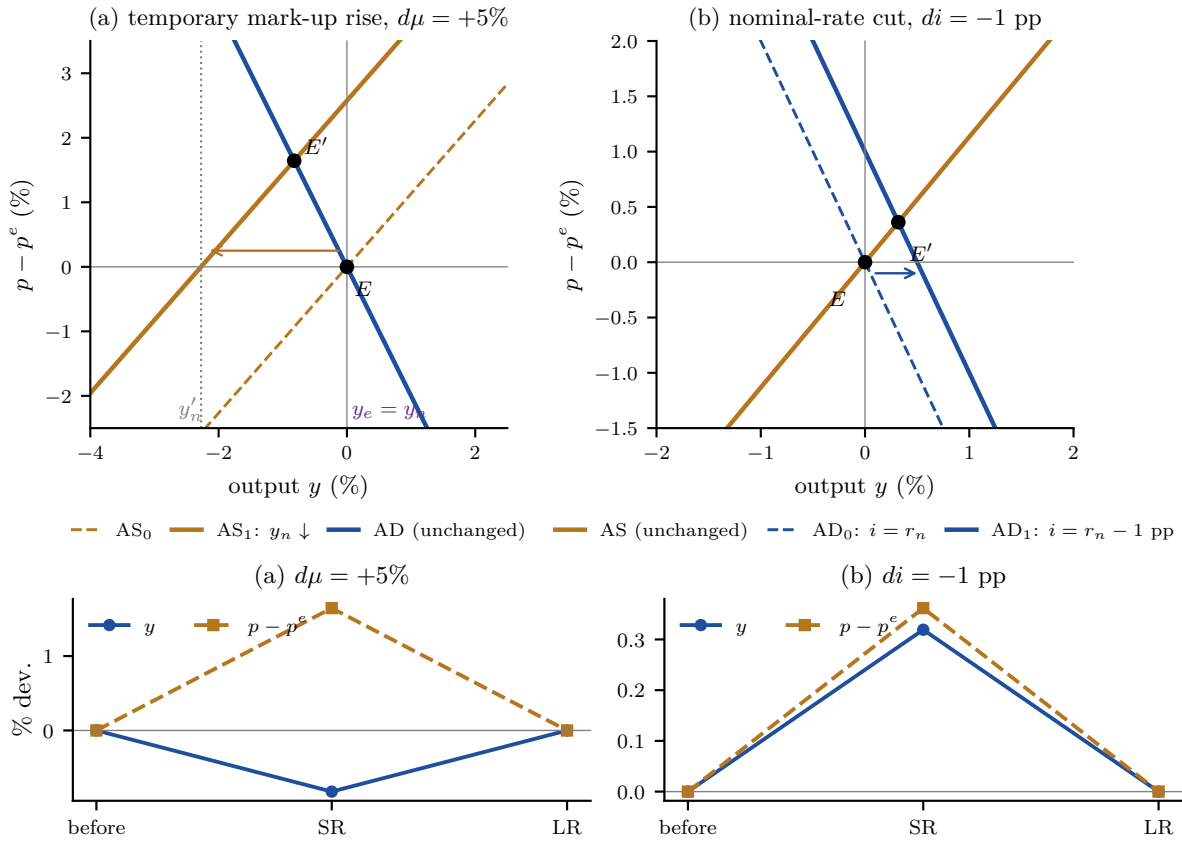
$i \downarrow \rightarrow r \downarrow$ ($\bar{p}$ anchored) $\rightarrow C \uparrow \rightarrow y \uparrow$ (sticky firms serve) $\rightarrow$ hours $\uparrow \rightarrow MC \uparrow \rightarrow$ adjusters raise $p \uparrow$ (partly offsets $r \downarrow$)

$$dy = -\frac{\sigma}{1 + \sigma\kappa} di > 0, \quad dp = \kappa dy > 0$$

$di = -1$ pp: $dy = +0,32\%$, $dp = +0,36\%$; $y - y_n = y - y_e = +0,32\%$ (overheating). Only $\frac{1}{1+\sigma\kappa} \approx 64\%$ of the AD shift reaches y .

LR: money neutral, only $\bar{p}$ could move.

No LM / liquidity effect: CB sets i directly. $\alpha \uparrow \rightarrow$ more y , less p .



(3) y_n vs y_e

- y_n : flexible prices, *decentralised* economy (with market power and taxes) $\rightarrow$ AS anchor, $y - y_n$ drives **prices**.
- y_e : *planner*, $\max u(C) - v(L)$ s.t. $Y = C + G$, $Y = AL$ $\rightarrow$ $y - y_e$ drives **welfare**.

Planner: $\frac{v_l}{u_c} = A$ (MRS = MRT). Market: $P = (1 + \mu) \frac{W}{A} \rightarrow \frac{v_l}{u_c} = \frac{W}{P} = \underbrace{\frac{A}{1 + \mu}}_{\text{wedge}}$

Logs: $\eta(y - a) + \sigma^{-1}(y - g) = a$ (planner) vs $= a - \mu$ (market) $\rightarrow$

$$y_n - y_e = -\frac{\mu}{\sigma^{-1} + \eta}$$

Why they differ: $P > MC \rightarrow W/P < MPL \rightarrow$ household works too little $\rightarrow Y_n < Y_e$ (lost surplus $\frac{1}{2}\mu^2/(\sigma^{-1} + \eta)$).

Why $\mu \uparrow$ moves y_n only: $y_n = y_n(a, g, \mu)$, $y_e = y_e(a, g)$ — the planner's problem has no prices, no market power, no taxes $\rightarrow \frac{dy_n}{d\mu} = -\frac{\mu}{\sigma^{-1} + \eta} < 0$, $dy_e = 0 \rightarrow$ inefficient shock $\rightarrow$ trade-off.

Contrast: $a \uparrow$ moves MRS and MRT in both problems $\rightarrow dy_n = dy_e$ (no trade-off).

$\# \alpha$ appears in neither: stickiness explains $y \neq y_n$, not $y_n \neq y_e$.

(4) $\min \mathcal{L} = \phi_y(y - y^*)^2 + \phi_p(p - p^e)^2$ s.t. $p - p^e = \kappa(y - y_n)$

Shape. Zero only at the bliss point (y^*, p^e) ; quadratic $\rightarrow$ symmetric (booms as bad as recessions: the CB *stabilises*, doesn't push y up) and convex (small misses cheap, large ones expensive) $\rightarrow$ spread the miss across both targets. Iso-loss = ellipses; ϕ_p/ϕ_y = conservatism (hawk vs dove). Only surprises $p - p^e$ cost (dispersion). Welfare version: $\phi_y = \frac{1}{2}$, $\phi_p = \frac{\theta}{2\kappa}$, $y^* = y_e$ (eq. 33).

Why subject to AS. One instrument i , and i is only in AD $\rightarrow$ moving i slides AD *along* AS $\rightarrow$ CB reaches every point of AS and only those $\rightarrow$ AS = feasible set (“budget line”). AD is not a constraint: it contains the free instrument; it only says which i implements the chosen point.

Solve (substitute AS into the loss): $\mathcal{L}(y) = \phi_y(y - y^*)^2 + \phi_p\kappa^2(y - y_n)^2$

$$\frac{d\mathcal{L}}{dy} = 0 \rightarrow \phi_y(y - y^*) + \kappa\phi_p(p - p^e) = 0 \quad (\text{targeting rule})$$

$$y^o = \frac{\phi_y y^* + \phi_p\kappa^2 y_n}{\phi_y + \phi_p\kappa^2}, \quad \mathcal{L}^o = \frac{\phi_y\phi_p\kappa^2}{\phi_y + \phi_p\kappa^2} (y^* - y_n)^2$$

Welfare weights $\rightarrow (y - y_e) + \theta(p - p^e) = 0$ (eq. 34); share of a mark-up shock into prices
 $= \frac{1}{1 + \theta\kappa} \approx 10\%$.

Policy meaning.

- $y^* = y_n$ (a, g shocks) $\rightarrow \mathcal{L}^o = 0, i = r_n$: divine coincidence.
- $y^* \neq y_n$ (μ shocks) $\rightarrow$ trade-off. $d\mu = 5\%$: $y^o = -2,05\%$, $p^o - p^e = +0,26\%$, $i^o = 5,84\%$ ($\uparrow$ from 2%, but $< r_n = 6,55\%$).
- CB picks a point *on* AS, cannot move AS $\rightarrow$ cannot close $y_n - y_e \rightarrow$ taxes / competition policy.
- Trade-off only SR and only with sticky prices ($\alpha \rightarrow 0 \rightarrow$ AS vertical). Can't hold $y > y_n$ every period: surprises can't be systematic (Kydland–Prescott).

Sign of the response depends on the mandate: IT steeper than AD $\rightarrow$ cut i .